

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

This session is not a board session. Both derivations are step-slide runs. The board carries **two ledger lines**, plus **one pair of empty axes** drawn at minute 12 and never wiped — every figure today is that picture redrawn, and the room should be able to look away from the screen and still see the axes.

Conventions. **ASK** — put it to the room before writing, and the answer to expect. **POINT** — the moments you turn and point at the board rather than the screen. **IF ASKED** — the question that always comes.

The board plan

Same room as sessions 3–6: blackboard across the front, screen down in the **centre**, a **left wing** and a **right wing**.

LEFT WING — the ledger. Two lines, written as they are derived, never wiped.

RIGHT WING — the axes at minute 12, and the vote tally at minute 9 above them. The tally is the only thing that gets wiped, and only at minute 26 when you resolve it.

Right wing — minute 9, the vote tally

Four options, four tick columns, and the count in the room's sight. Write the distribution down. Do **not** comment on it, do **not** hint, and do **not** resolve it — the resolution is step 7 of derivation 1 at minute 26, and the whole shape of the first half depends on the question staying open for seventeen minutes.

IF ASKED *which is right?* — "You are going to derive it. If I tell you now, you will believe me instead of the algebra."

Right wing — minute 12, the axes

Draw them once, large, and leave them:

- horizontal: **protein p**
- vertical: **rate (molecules per minute)**

Then two curves, in this order and no others: the **flat production line** and the **straight removal line through the origin**. Nothing else goes on these axes for the rest of the session; when you need a new picture, gesture at them and let the slide carry the detail.

ASK while drawing removal: *why is removal a straight line through the origin and not a constant?* Expect the answer from session 5 — dilution removes a **fraction** per minute, so the amount removed grows with the pool. If nobody has it, ask what happens to removal when $p = 0$.

DO NOT WIPE. You point at these axes at 32, at 50, at 69 and at 84.

Line 1 — at minute 24, step 6 of derivation 1

Left wing, and **box it**.

speed-up $\lambda \implies$ cost λ , forever (when λ comes from γ)

POINT at the axes as you write it. The tag moves the **removal** line — it tilts it steeper — and the crossing slides down, so α has to rise to put it back. Both effects are the same λ because both live in the same denominator.

ASK: *is that an approximation?* No. It is exact, and it is exact for the same reason it is boring: one denominator, two requirements.

Line 2 — at minute 43, the last step of derivation 2

Directly under line 1.

NAR — α raised, γ untouched $\implies$ steady-state flux unchanged

Then turn to the axes. This is the moment the session turns:

The removal line never moved. The crossing is where the two curves agree, so the rate at the crossing is whatever removal says it is — and removal has not heard about your circuit.

ASK: *so negative autoregulation is free?* No — it is **prepaid**. The transient costs more, the steady state costs the same, and which of those dominates depends entirely on how long you run. That is item 4 of the handout and the second rhythm at 50 min.

IF ASKED *does NAR really not change γ ?* — not by itself. If you build autoregulation by wiring in a protease adapter, you have built both things, and then you pay both bills. Good question, and worth thirty seconds.

What does not go on the board

- Either derivation. They are step-slide runs; that is the point of them.
 - The Rosenfeld numbers. They are on the slide and in the paper.
 - The bistability picture. It is a figure with three curves on it and you will make a mess of it in chalk.
 - Anything during the handout. Circulate instead.
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If you are running late

Cut in this order:

1. **13c, what positive autoregulation buys you, at 74 min** — down to two minutes: keep "you have built a memory" and the growth-rate caveat, drop the knob list, which is on the ledger slide at 81 anyway. **This is the cheapest cut in the session:** it is the only one whose content survives somewhere else in the same hour.
2. **9a, the Rosenfeld design slide at 44 min** — merge into 9b: put the figure up, take the axes from the room in thirty seconds, and go straight to the result.
3. **The variance segment at 77 min** — down to one minute: say NAR narrows the distribution as well as speeding it up, that the mechanism is the same restoring gap acting on fluctuations, and that the derivation is session 12. **Cut this one last, and read the note below before you do.**
4. **Nothing else.** Do not cut the handout and do not cut the design ledger.

***Why variance moved from first cut to last, 14 September.** T17's assessment moved off PS3 and onto PS6 that day, so nothing between now and session 12 asks a student to think about variance again — the four minutes at 77 are the only demonstration it gets, and session 12's worked comparison is an obligation on a deck that does not yet exist. When this cut order was written, PS3 was going to make them write about the mechanism within a week and one minute here was cheap. It is not cheap now. The knob list at 74 is duplicated on the ledger; variance is duplicated nowhere.*

Never cut: the vote resolution at 26 min and line 2 at 43. Those two together are the session, and session 9's toggle has no foundation without the second half of the picture.

The closing question for Tuesday

Last surface, and leave it up while they pack:

Two proteins, each repressing the other. Draw that picture.

They cannot — not on one pair of axes — and that is the cold open for session 8.